

Day 15 Weekly Assessment

Geometry Summer Credit | Week 3 Cumulative Check | 25 Minutes

Name:	Date:	Score:
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Directions

This replaces today's daily quiz. Show equations for short-response items. Round decimals to the nearest tenth. Law of Sines and Law of Cosines are not assessed.

1.	A right triangle has legs 6 and 8. Its hypotenuse is: A 7 B 10 C 12 D 14
2.	Which set of side lengths forms a right triangle? A 5, 6, 7 B 6, 8, 10 C 7, 8, 12 D 8, 9, 10
3.	A 45-45-90 triangle has leg 7. Its hypotenuse is: A 7 B $7\sqrt{2}$ C 14 D $14\sqrt{2}$
4.	A 30-60-90 triangle has hypotenuse 20. Its short leg is: A 5 B 10 C $10\sqrt{3}$ D $20\sqrt{3}$
5.	A right triangle has angle 40 degrees and hypotenuse 15. Which equation finds the opposite side x? A $x=15\sin(40)$ B $x=15\cos(40)$ C $x=15\tan(40)$ D $x=15/\sin(40)$
6.	Which formula finds the area when sides a and b and their included angle C are known? A $A=ab$ B $A=(1/2)ab$ C $A=(1/2)ab \sin(C)$ D $A=a^2+b^2$
7.	Simplify $\sqrt{180}$.

Day 15 Weekly Assessment

Continued | [Show Complete Setups](#)

8.	Find the distance between $(1, -2)$ and $(7, 6)$.
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9.	A right triangle has opposite side 10 and adjacent side 14. Find the acute reference angle.
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10.	A person stands 30 ft from a pole. The angle of elevation to the top is 47 degrees. The person's eye level is 5 ft. Find the pole height.
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11.	Find the area of a triangle with sides 12 cm and 19 cm and included angle 55 degrees.
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12.	Error analysis: A student solves a 30-60-90 triangle with short leg 9 and says the hypotenuse is $9\sqrt{3}$. Explain and correct the error.
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